

Q1. In an experiment of measuring the refractive index of a glass slab using travelling microscope in physics lab, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of the glass slab at 5.00 mm. Travelling microscope has 20 divisions in one cm on main scale and 50 divisions on Vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10} \times 10^{-3}$, where x is _____

Q2. The equation of a circle is given by $x^2 + y^2 = a^2$, where a is the radius. If the equation is modified to change the origin other than $(0, 0)$, then find out the correct dimensions of A and B in a new equation :

$$(x - At)^2 + (y - \frac{t}{B})^2 = a^2$$

The dimensions of t is given as $[T^{-1}]$

(1) $A = [L^{-1}T]$, $B = [LT^{-1}]$

(2) $A = [LT]$, $B = [L^{-1}T^{-1}]$

(3) $A = [L^{-1}T^{-1}]$, $B = [LT^{-1}]$

(4) $A = [L^{-1}T^{-1}]$, $B = [LT]$

Q3. An object moves at a constant speed along a circular path in a horizontal plane with centre at the origin. When the object is at $x = +2$ m, its velocity is $-4\hat{j} \text{ m s}^{-1}$. The object's velocity (v) and acceleration (a) at $x = -2$ m will be

(1) $v = 4\hat{i} \text{ m s}^{-1}$, $a = 8\hat{j} \text{ m s}^{-2}$

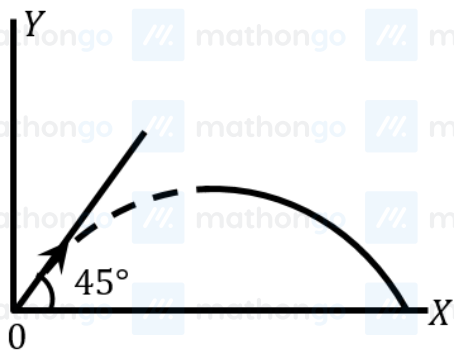
(2) $v = 4\hat{j} \text{ m s}^{-1}$, $a = 8\hat{i} \text{ m s}^{-2}$

(3) $v = -4\hat{j} \text{ m s}^{-1}$, $a = 8\hat{i} \text{ m s}^{-2}$

(4) $v = -4\hat{i} \text{ m s}^{-1}$, $a = -8\hat{j} \text{ m s}^{-2}$

Q4. A car is moving on a circular path of radius 600 m such that the magnitudes of the tangential acceleration and centripetal acceleration are equal. The time taken by the car to complete first quarter of revolution, if it is moving with an initial speed of 54 km h^{-1} is $t(1 - e^{-\frac{\pi}{2}})$ s. The value of t is _____.

Q5. A particle of mass 100 g is projected at time $t = 0$ with a speed 20 m s^{-1} at an angle 45° to the horizontal as given in the figure. The magnitude of the angular momentum of the particle about the starting point at time $t = 2$ s is found to be $\sqrt{K} \text{ kg m}^2 \text{ s}^{-1}$. The value of K is _____.
(Take $g = 10 \text{ m s}^{-2}$)



Q6. The time taken by an object to slide down 45° rough inclined plane is n times as it takes to slide down a perfectly smooth 45° incline plane. The coefficient of kinetic friction between the object and the incline plane is:

(1) $\sqrt{\frac{1}{1-n^2}}$

(2) $\sqrt{1 - \frac{1}{n^2}}$

(3) $1 + \frac{1}{n^2}$

(4) $1 - \frac{1}{n^2}$

Q7. Force acts for 20 s on a body of mass 20 kg, starting from rest, after which the force ceases and then body describes 50 m in the next 10 s. The value of force will be :

- (1) 40 N (2) 5 N
(3) 20 N (4) 10 N

Q8. Identify the correct statements from the following:

- (A) Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket is negative
(B) Work done by gravitational force in lifting a bucket out of a well by a rope tied to the bucket is negative
(C) Work done by friction on a body sliding down an inclined plane is positive
(D) Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity is zero
(E) Work done by the air resistance on an oscillating pendulum is negative

Choose the correct answer from the options given below:

- (1) B and E only (2) A and C only
(3) B, D and E only (4) B and D only

Q9. The time period of a satellite of earth is 24 hours. If the separation between the earth and the satellite is decreased to one fourth of the previous value, then its new time period will become.

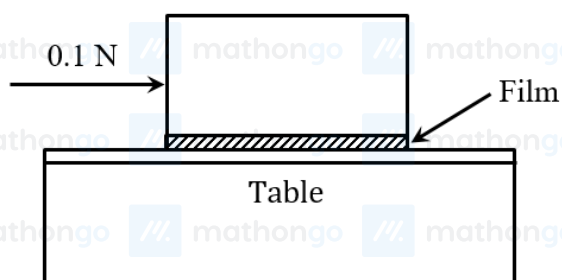
- (1) 4 hours (2) 6 hours
(3) 12 hours (4) 3 hours

Q10. A fully loaded boeing aircraft has a mass of 5.4×10^5 kg. Its total wing area is 500 m^2 . It is in level flight with a speed of 1080 km h^{-1} . If the density of air ρ is 1.2 kg m^{-3} , the fractional increase in the speed of the air on the upper surface of the wing relative to the lower surface in percentage will be

($g = 10 \text{ m s}^{-2}$)

- (1) 16 (2) 6
(3) 8 (4) 10

Q11. A metal block of base area 0.20 m^2 is placed on a table, as shown in figure. A liquid film of thickness 0.25 mm is inserted between the block and the table. The block is pushed by a horizontal force of 0.1 N and moves with a constant speed. If the viscosity of the liquid is $5.0 \times 10^{-3} \text{ Pl}$, the speed of block is $\times 10^{-3} \text{ m s}^{-1}$.



Q12. Heat energy of 184 kJ is given to ice of mass 600 g at -12°C , Specific heat of ice is $2222.3 \text{ J kg}^{-1} ^\circ \text{C}^{-1}$ and latent heat of ice is 336 kJ kg^{-1} .

- (A) Final temperature of system will be 0°C
(B) Final temperature of the system will be greater than 0°C
(C) The final system will have a mixture of ice and water in the ratio of $5 : 1$
(D) The final system will have a mixture of ice and water in the ratio of $1 : 5$
(E) The final system will have water only

Choose the correct answer from the options given below :

- (1) A and D only
(3) A and E only
- (2) B and D only
(4) A and C only

Q13. At 300 K, the rms speed of oxygen molecules is $\sqrt{\frac{\alpha+5}{\alpha}}$ times to that of its average speed in the gas. Then, the value of α will be (use $\pi = \frac{22}{7}$)

- (1) 32
(3) 24
- (2) 28
(4) 27

Q14. A particle of mass 250 g executes a simple harmonic motion under a periodic force $F = (-25x)$ N. The particle attains a maximum speed of 4 m s^{-1} during its oscillation. The amplitude of the motion is _____ cm.

Q15. A point charge $2 \times 10^{-2} \text{ C}$ is moved from P to S in a uniform electric field of 30 N C^{-1} directed along positive x -axis. If coordinates of P and S are $(1, 2, 0) \text{ m}$ and $(0, 0, 0) \text{ m}$ respectively, the work done by electric field will be

- (1) 1200 mJ
(3) -600 mJ
- (2) 600 mJ
(4) -1200 mJ

Q16. For a charged spherical ball, electrostatic potential inside the ball varies with r as $V = 2ar^2 + b$.

Here, a and b are constant and r is the distance from the center. The volume charge density inside the ball is $-\lambda a \epsilon$. The value of λ is _____.

ϵ = permittivity of medium.

Q17. With the help of potentiometer, we can determine the value of emf of a given cell. The sensitivity of the potentiometer is

- (A) directly proportional to the length of the potentiometer wire
(B) directly proportional to the potential gradient of the wire
(C) inversely proportional to the potential gradient of the wire
(D) inversely proportional to the length of the potentiometer wire

Choose the correct option for the above statements:

- (1) B and D only
(3) A only
- (2) A and C only
(4) C only

Q18. A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by 5Ω . When a resistance of 15Ω is used for shunting null point moves to 300 cm. The internal resistance of the cell is _____ Ω .

Q19. The electric current in a circular coil of four turns produces a magnetic induction 32 T at its centre. The coil is unwound and is rewound into a circular coil of single turn, the magnetic induction at the centre of the coil by the same current will be :

- (1) 8 T
(3) 2 T
- (2) 4 T
(4) 16 T

Q20. A square loop of area 25 cm^2 has a resistance of 10Ω . The loop is placed in uniform magnetic field of magnitude 40.0 T. The plane of loop is perpendicular to the magnetic field. The work done in pulling the loop out of the magnetic field slowly and uniformly in 1.0 sec, will be

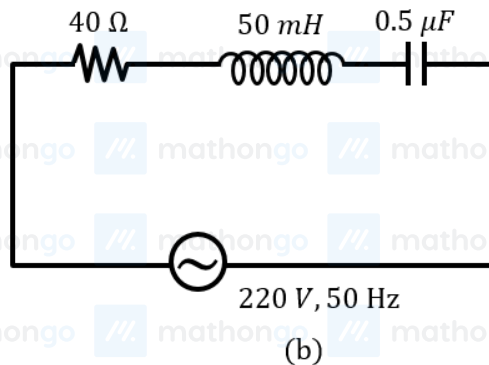
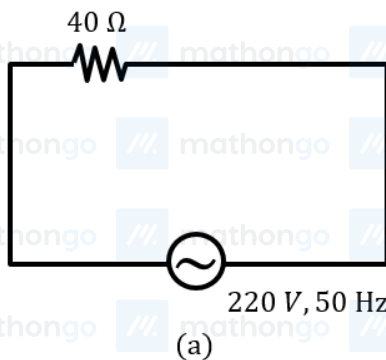
(1) $2.5 \times 10^{-3} \text{ J}$

(3) $1.0 \times 10^{-4} \text{ J}$

(2) $1.0 \times 10^{-3} \text{ J}$

(4) $5 \times 10^{-3} \text{ J}$

Q21. For the given figures, choose the correct options:



- (1) The rms current in circuit (b) can never be larger than that in (a) (2) The rms current in figure (a) is always equal to that in figure (b)
- (3) The rms current in circuit (b) can be larger than that in (a) (4) At resonance, current in (b) is less than that in (a)

Q22. An inductor of inductance $2 \mu\text{H}$ is connected in series with a resistance, a variable capacitor and an AC source of frequency 7 kHz . The value of capacitance for which maximum current is drawn into the circuit is $\frac{1}{x} \text{ F}$, where the value of x is _____.
(Take $\pi = \frac{22}{7}$)

Q23. Given below are two statements :

Statement I : Electromagnetic waves are not deflected by electric and magnetic field.

Statement II : The amplitude of electric field and the magnetic field in electromagnetic waves are related to each other as $E_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} B_0$.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Statement I is true but statement II is false (2) Both Statement I and Statement II are true
- (3) Statement I is false but statement II is true (4) Both Statement I and Statement II are false

Q24. A scientist is observing a bacteria through a compound microscope. For better analysis and to improve its resolving power he should. (Select the best option)

- (1) Increase the wave length of the light (2) Increase the refractive index of the medium between the object and objective lens
- (3) Decrease the focal length of the eye piece (4) Decrease the diameter of the objective lens

Q25. Unpolarised light is incident on the boundary between two dielectric media, whose dielectric constants are 2.8 (medium –1) and 6.8 (medium – 2), respectively. To satisfy the condition, so that the reflected and refracted rays are perpendicular to each other, the angle of incidence should be $\tan^{-1} \left(1 + \frac{10}{\theta}\right)^{\frac{1}{2}}$, the value of θ is _____.
(Given for dielectric media, $\mu_r = 1$)

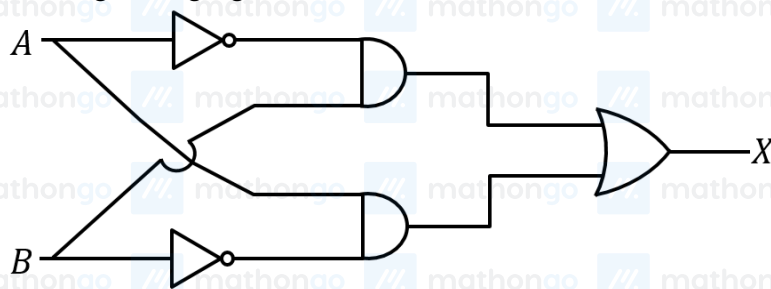
Q26. The ratio of de-Broglie wavelength of an α -particle and a proton accelerated from rest by the same potential is $\frac{1}{\sqrt{m}}$, the value of m is:

- (1) 4 (2) 16
(3) 8 (4) 2

Q27. Substance A has atomic mass number 16 and half life of 1 day. Another substance B has atomic mass number 32 and half life of $\frac{1}{2}$ day. If both A and B simultaneously start undergo radio activity at the same time with initial mass 320 g each, how many total atoms of A and B combined would be left after 2 days

- (1) 3.38×10^{24} (2) 6.76×10^{24}
(3) 6.76×10^{23} (4) 1.69×10^{24}

Q28. For the given logic gates combination, the correct truth table will be



(1) A B X

0	0	1
0	1	0
1	0	0
1	1	0

(3) A B X

0	0	1
0	1	0
1	0	1
1	1	0

(2) A B X

0	0	0
0	1	1
1	0	1
1	1	0

(4) A B X

0	0	0
0	1	1
1	0	1
1	1	1

Q29. The modulation index for an A.M. wave having maximum and minimum peak to peak voltages of 14 mV and 6 mV respectively is :

- (1) 1.4 (2) 0.4
(3) 0.2 (4) 0.6

Q30. When two resistance R_1 and R_2 connected in series and introduced into the left gap of a meter bridge and a resistance of 10Ω is introduced into the right gap, a null point is found at 60 cm from left side. When R_1 and R_2 are connected in parallel and introduced into the left gap, a resistance of 3Ω is introduced into the right-gap to get null point at 40 cm from left end. The product of $R_1 R_2$ is _____ Ω^2

Q31. Assume that the radius of the first Bohr orbit of hydrogen atom is $0.6 \text{ \AA}$. The radius of the third Bohr orbit of He^+ is _____ picometer. (Nearest Integer)

Q32. Given below are two statements:

Statement I: The decrease in first ionization enthalpy from B to Al is much larger than that from Al to Ga.

Statement II: The d orbitals in Ga are completely filled.

In the light of the above statements, choose the most appropriate answer from the options given below

- (1) Statement I is incorrect but statement II is correct.
 (2) Both the statements I and II are correct
 (3) Statement I is correct but statement II is incorrect
 (4) Both the statements I and II are incorrect

Q33. According to MO theory the bond orders for O_2^{2-} , CO and NO^+ respectively, are

- (1) 1, 3 and 3
 (2) 1, 3 and 2
 (3) 1, 2 and 3
 (4) 2, 3 and 3

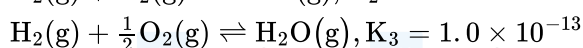
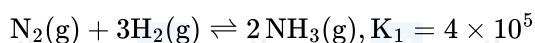
Q34. Which of the following relations are correct?

- (A) $\Delta U = q + p\Delta V$
 (B) $\Delta G = \Delta H - T\Delta S$
 (C) $\Delta S = \frac{q_{rev}}{T}$
 (D) $\Delta H = \Delta U - \Delta nRT$

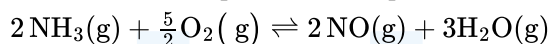
Choose the most appropriate answer from the options given below :

- (1) C and D only
 (2) B and C only
 (3) A and B only
 (4) B and D only

Q35. At 298 K



Based on above equilibria, the equilibrium constant of the reaction,



is $\times 10^{-33}$ (Nearest integer)

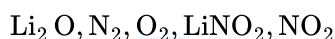
Q36. The volume of HCl, containing $73g L^{-1}$, required to completely neutralise NaOH obtained by reacting 0.69 g of metallic sodium with water, is _____ mL. (Nearest Integer)

(Given : molar Masses of Na, Cl, O, H are 23, 35.5, 16 and $1g mol^{-1}$ respectively)

Q37. An indicator 'X' is used for studying the effect of variation in concentration of iodide on the rate of reaction of iodide ion with H_2O_2 at room temp. The indicator 'X' forms blue colored complex with compound 'A' present in the solution. The indicator 'X' and compound 'A' respectively are

- (1) Starch and iodine
 (2) Methyl orange and H_2O_2
 (3) Starch and H_2O_2
 (4) Methyl orange and iodine

Q38. On heating, $LiNO_3$ gives how many compounds among the following?



Q39. Given below are two statements:

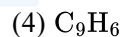
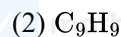
Statement I : Nickel is being used as the catalyst for producing syn gas and edible fats.

Statement II : Silicon forms both electron rich and electron deficient hydrides.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both the statements I and II are correct
 (2) Statement I is incorrect but statement II is correct
 (3) Both the statements I and II are incorrect
 (4) Statement I is correct but statement II is incorrect

Q40. When a hydrocarbon A undergoes combustion in the presence of air, it requires 9.5 equivalents of oxygen and produces 3 equivalents of water. What is the molecular formula of A?



Q41. When 0.01 mol of an organic compound containing 60% carbon was burnt completely, 4.4 g of CO_2 was produced. The molar mass of compound is _____ $gmol^{-1}$ (Nearest integer)

Q42. The concentration of dissolved Oxygen in water for growth of fish should be more than X ppm and Biochemical Oxygen Demand in clean water should be less than Y ppm. X and Y in ppm are, respectively.

(1) X Y

(2) X Y

6 5

4 8

(3) X Y

(4) X Y

4 15

6 12

Q43. A metal M forms hexagonal close-packed structure. The total number of voids in 0.02 mol of it is _____ $\times 10^{21}$ (Nearest integer)
(Given $N_A = 6.02 \times 10^{23}$)

Q44. Match List I with List II.

List I

List II

A. van't Hoff factor, i

I. Cryoscopic constant

B. k_f

II. Isotonic solutions

C. Solutions with same osmotic pressure

III. $\frac{\text{Normal molar mass}}{\text{Abnormal molar mas}}$

D. Azeotropes

IV. Solutions with same composition of vapour above it

Choose the correct answer from the options given below:

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-II, C-I, D-IV

(3) A-III, B-I, C-IV, D-II

(4) A-I, B-III, C-II, D-IV

Q45. The equilibrium constant for the reaction $Zn(s) + Sn^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Sn(s)$ is 1×10^{20} at 298 K. The magnitude of standard electrode potential of Sn/Sn^{2+} if $E_{Zn^{2+}/Zn}^0 = -0.76$ V is _____ $\times 10^{-2}$ V. (Nearest integer)

Given : $\frac{2.303RT}{F} = 0.059$ V

Q46. For conversion of compound $A \rightarrow B$, the rate constant of the reaction was found to be $4.6 \times 10^{-5} \text{ Lmol}^{-1} \text{ s}^{-1}$. The order of the reaction is _____.

Q47. Match List-I and List-II

List-I

List-II

A. Osmosis

I. Solvent molecules pass through semi permeable membrane towards solvent side.

B. Reverse osmosis

Movement of charged colloidal particles under the influence of applied electric potential towards oppositely charged electrodes

C. Electro osmosis III. Solvent molecules pass through semi permeable membrane towards solution side

D. Electrophoresis IV. Dispersion medium moves in an electric field.

Choose the correct answer from the options given below:

(1) A-I, B-III, C-IV, D-II

(2) A-III, B-I, C-IV, D-II

(3) A-III, B-I, C-II, D-IV

(4) A-I, B-III, C-II, D-IV

Q48. The major component of which of the following ore is sulphide based mineral?

(1) Calamine

(2) Siderite

(3) Sphalerite

(4) Malachite

Q49. Total number of acidic oxides among N_2O_3 , NO_2 , N_2O , Cl_2O_7 , SO_2 , CO , CaO , Na_2O and NO is _____.

Q50. A solution of CrO_5 in amyl alcohol has a....colour

(1) Green

(2) Orange-Red

(3) Yellow

(4) Blue

Q51. The set of correct statements is:

(i) Manganese exhibits +7 oxidation state in its oxide.

(ii) Ruthenium and Osmium exhibit +8 oxidation in their oxides.

(iii) Sc shows +4 oxidation state which is oxidizing in nature.

(iv) Cr shows oxidising nature in +6 oxidation state.

(1) (ii) and (iii)

(2) (i), (ii) and (iv)

(3) (i) and (iii)

(4) (ii), (iii) and (iv)

Q52. Correct order of spin only magnetic moment of the following complex ions is:

(Given At. No. Fe : 26, Co : 27)

(1) $[\text{FeF}_6]^{3-} > [\text{CoF}_6]^{3-} > [\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$

(2) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-} > [\text{CoF}_6]^{3-} > [\text{FeF}_6]^{3-}$

(3) $[\text{FeF}_6]^{3-} > [\text{Co}(\text{C}_2\text{O}_4)_3]^{3-} > [\text{CoF}_6]^{3-}$

(4) $[\text{CoF}_6]^{3-} > [\text{FeF}_6]^{3-} > [\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$

Q53. The denticity of the ligand present in the Fehling's reagent is _____.

Q54. The one giving maximum number of isomeric alkenes on dehydrohalogenation reaction is (excluding rearrangement)

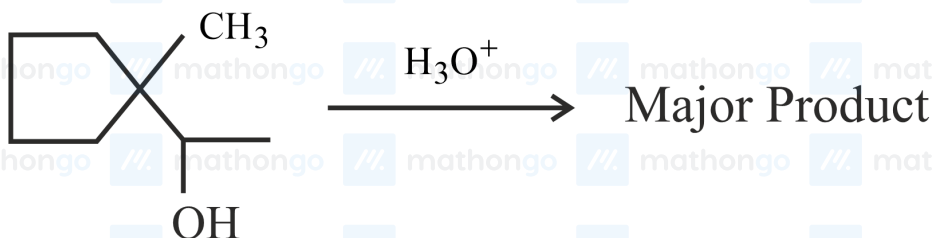
(1) 1-Bromo-2-methylbutane

(2) 2-Bromopropane

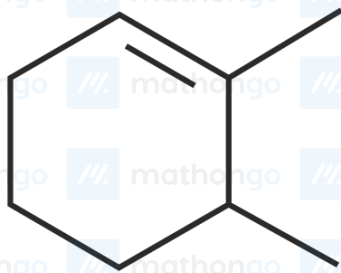
(3) 2-Bromopentane

(4) 2-Bromo-3, 3-dimethylpentane

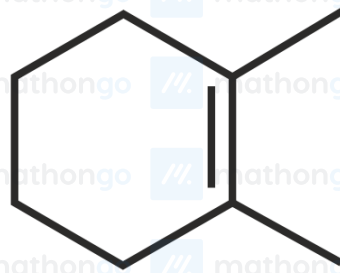
Q55. Find out the major product for the following reaction



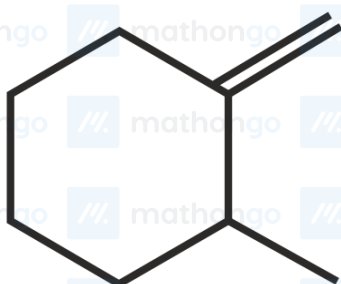
(1)



(2)



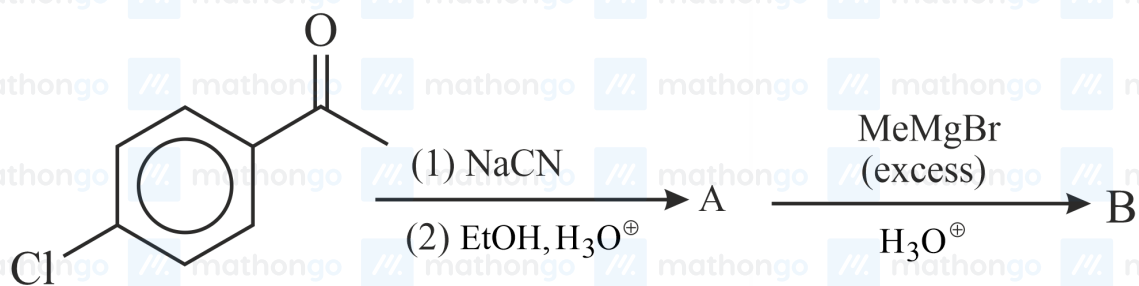
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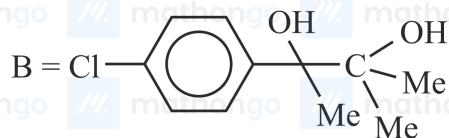
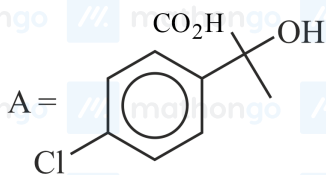
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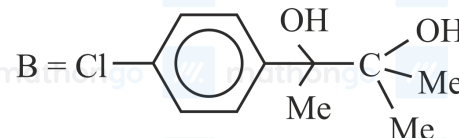
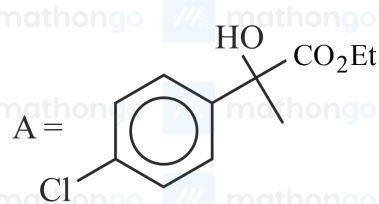
Q56. Find out the major products from the following reaction sequence.



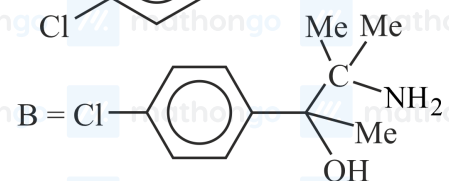
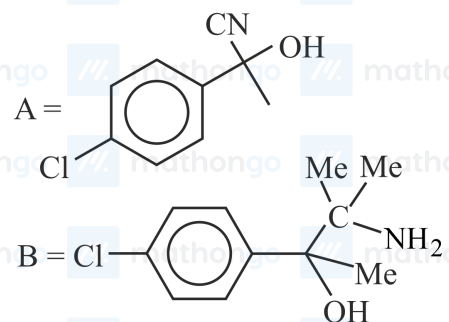
(1)



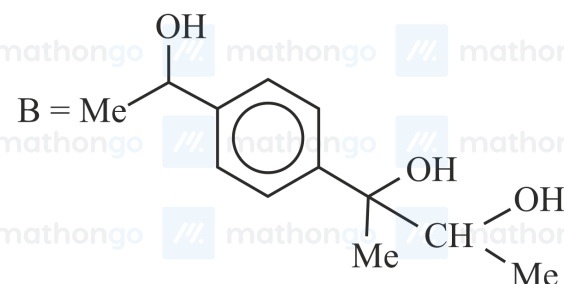
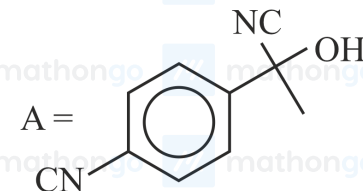
(2)



(3)



(4)



Q57. Reaction of propanamide with Br_2 / KOH (aq) produces:

(1) Ethylnitrile

(2) Propylamine

(3) Propanenitrile

(4) Ethylamine

Q58. Match List-I and List-II.

List-I

A. Elastomeric polymer

B. Fibre polymer

C. Thermosetting polymer

D. Thermoplastic polymer

List-II

I. Urea formaldehyde resin

II. Polystyrene

III. Polyester

IV. Neoprene

Choose the correct answer from the options given below:

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-IV, B-III, C-I, D-II

(4) A-IV, B-I, C-III, D-II

Q59. A doctor prescribed the drug Equanil to a patient. The patient was likely to have symptoms of which disease?

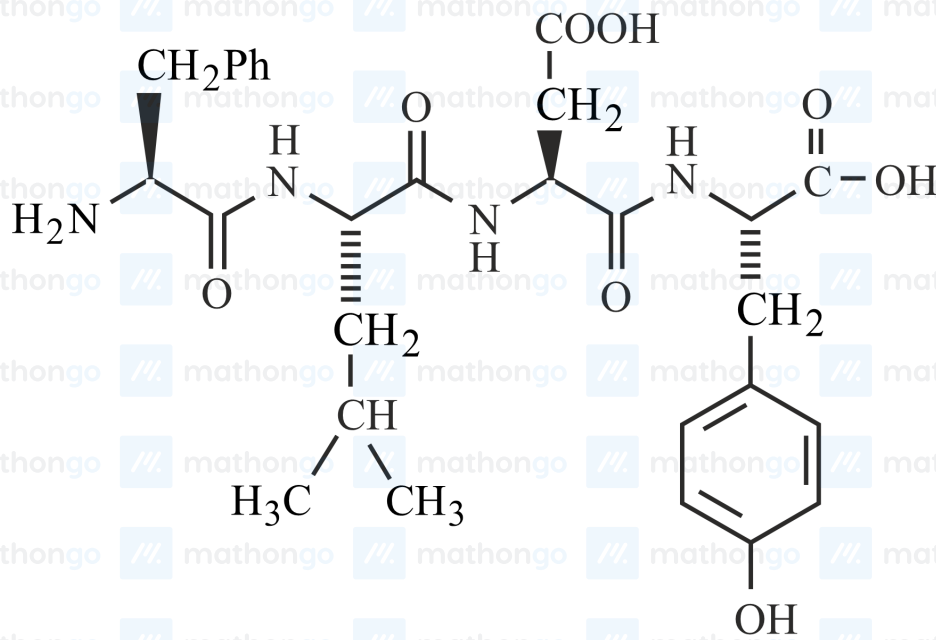
(1) Stomach ulcers

(2) Hyperacidity

(3) Anxiety and stress

(4) Depression and hypertension

Q60. Following tetrapeptide can be represented as



(F, L, D, Y, I, Q, P are one letter codes for amino acids)

(1) FIQY

(2) FLDY

(3) YQLF

(4) PLDY

Q61. Let $\alpha_1, \alpha_2, \dots, \alpha_7$ be the roots of the equation $x^7 + 3x^5 - 13x^3 - 15x = 0$ and

$|\alpha_1| \geq |\alpha_2| \geq \dots \geq |\alpha_7|$.

Then, $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ is equal to _____

Q62. Let $\alpha = 8 - 14i$, $A = \left\{ z \in \mathbb{C} : \frac{\alpha z - \bar{\alpha} \bar{z}}{z^2 - (\bar{z})^2 - 112i} = 1 \right\}$ and $B = \{ z \in \mathbb{C} : |z + 3i| = 4 \}$

Then, $\sum_{z \in A \cap B} (Re\ z - Im\ z)$ is equal to _____

Q63. The letters of the word OUGHT are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word TOUGH is :

(1) 89

(2) 84

(3) 86

(4) 79

Q64. The total number of 4-digit numbers whose greatest common divisor with 54 is 2, is

Q65. Let $a_1 = b_1 = 1$ and $a_n = a_{n-1} + (n-1)$, $b_n = b_{n-1} + a_{n-1}$, $\forall n \geq 2$. If $S = \sum_{n=1}^{10} \left(\frac{b_n}{2^n} \right)$ and

$T = \sum_{n=1}^8 \frac{n}{2^{n-1}}$ then $2^7(2S - T)$ is equal to

Q66. Let $\{a_k\}$ and $\{b_k\}$, $k \in \mathbb{N}$, be two G.P.s with common ratio r_1 and r_2 respectively such that $a_1 = b_1 = 4$ and $r_1 < r_2$. Let $c_k = a_k + b_k$, $k \in \mathbb{N}$. If $c_2 = 5$ and $c_3 = \frac{13}{4}$ then $\sum_{k=1}^{\infty} c_k - (12a_6 + 8b_4)$ is equal to

Q67. The number of 3 digit numbers, that are divisible by either 3 or 4 but not divisible by 48, is

(1) 472

(2) 432

(3) 507

(4) 400

Q68. Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1+x)^{99}$. Let a be the middle term in the expansion of $\left(2 + \frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{{}^{200}C_{99}K}{a} = \frac{2^l m}{n}$, where m and n are odd numbers, then the ordered pair (l, n) is equal to:

- (1) (50, 51) (2) (51, 99)
(3) (50, 101) (4) (51, 101)

Q69. The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$

- (1) $[-2, -1]$ (2) $[-2, -\frac{3}{2}]$
(3) $[-1, -\frac{1}{2}]$ (4) $[-\frac{3}{2}, -1]$

Q70. A circle with centre $(2, 3)$ and radius 4 intersects the line $x + y = 3$ at the points P and Q . If the tangents at P and Q intersect at the point $S(\alpha, \beta)$, then $4\alpha - 7\beta$ is equal to

Q71. A triangle is formed by the tangents at the point $(2, 2)$ on the curves $y^2 = 2x$ and $x^2 + y^2 = 4x$, and the line $x + y + 2 = 0$. If r is the radius of its circumcircle, then r^2 is equal to

Q72. If the tangent at a point P on the parabola $y^2 = 3x$ is parallel to the line $x + 2y = 1$ and the tangents at the points Q and R on the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ are perpendicular to the line $x - y = 2$, then the area of the triangle PQR is:

- (1) $\frac{9}{\sqrt{5}}$ (2) $5\sqrt{3}$
(3) $\frac{3}{2}\sqrt{5}$ (4) $3\sqrt{5}$

Q73. The statement $B \Rightarrow ((\sim A) \vee B)$ is not equivalent to :

- (1) $B \Rightarrow (A \Rightarrow B)$ (2) $A \Rightarrow (A \Leftrightarrow B)$
(3) $A \Rightarrow ((\sim A) \Rightarrow B)$ (4) $B \Rightarrow ((\sim A) \Rightarrow B)$

Q74. Let $X = \{11, 12, 13, \dots, 40, 41\}$ and $Y = \{61, 62, 63, \dots, 90, 91\}$ be the two sets of observations. If $\bar{x}$ and $\bar{y}$ are their respective means and σ^2 is the variance of all the observations in $X \cup Y$, then $|\bar{x} + \bar{y} - \sigma^2|$ is equal to

Q75. Let R be a relation defined on $\mathbb{N}$ as $a R b$ is $2a + 3b$ is a multiple of 5, $a, b \in \mathbb{N}$. Then R is

- (1) not reflexive (2) transitive but not symmetric
(3) symmetric but not transitive (4) an equivalence relation

Q76.

The set of all values of $t \in \mathbb{R}$, for which the matrix $\begin{bmatrix} e^t & e^{-t}(\sin t - 2\cos t) & e^{-t}(-2\sin t - \cos t) \\ e^t & e^{-t}(2\sin t + \cos t) & e^{-t}(\sin t - 2\cos t) \\ e^t & e^{-t}\cos t & e^{-t}\sin t \end{bmatrix}$

is invertible, is

- (1) $\{(2k+1)\frac{\pi}{2}, k \in \mathbb{Z}\}$ (2) $\{k\pi + \frac{\pi}{4}, k \in \mathbb{Z}\}$
(3) $\{k\pi, k \in \mathbb{Z}\}$ (4) $\mathbb{R}$

Q77. Let A be a symmetric matrix such that $|A| = 2$ and $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$. If the sum of the diagonal elements of A is s , then $\frac{\beta s}{\alpha^2}$ is equal to _____.

Q78. Consider a function $f: \mathbb{N} \rightarrow \mathbb{R}$, satisfying $f(1) + 2f(2) + 3f(3) + \dots + xf(x) = x(x+1)f(x); x \geq 2$ with $f(1) = 1$. Then $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ is equal to

(1) 8200

(3) 8400

(2) 8000

(4) 8100

Q79. Let f and g be twice differentiable functions on R such that

$$f''(x) = g''(x) + 6x$$

$$f'(1) = 4g'(1) - 3 = 9$$

$$f(2) = 3g(2) = 12$$

Then which of the following is NOT true ?

$$(1) g(-2) - f(-2) = 20$$

$$(2) \text{ If } -1 < x < 2, \text{ then } |f(x) - g(x)| < 8$$

$$(3) |f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$$

$$(4) \text{ There exists } x_0 \in (1, \frac{3}{2}) \text{ such that}$$

$$f(x_0) = g(x_0)$$

Q80. If the equation of the normal to the curve $y = \frac{x-a}{(x+b)(x-2)}$ at the point $(1, -3)$ is $x - 4y = 13$ then the value of $a + b$ is equal to _____

Q81. The value of the integral $\int_1^2 \left(\frac{t^4+1}{t^6+1} \right) dt$ is :

$$(1) \tan^{-1} \frac{1}{2} + \frac{1}{3} \tan^{-1} 8 - \frac{\pi}{3}$$

$$(2) \tan^{-1} 2 - \frac{1}{3} \tan^{-1} 8 + \frac{\pi}{3}$$

$$(3) \tan^{-1} 2 + \frac{1}{3} \tan^{-1} 8 - \frac{\pi}{3}$$

$$(4) \tan^{-1} \frac{1}{2} - \frac{1}{3} \tan^{-1} 8 + \frac{\pi}{3}$$

Q82. The value of the integral $\int_{1/2}^2 \frac{\tan^{-1} x}{x} dx$ is equal to

$$(1) \pi \log_e 2$$

$$(2) \frac{1}{2} \log_e 2$$

$$(3) \frac{\pi}{4} \log_e 2$$

$$(4) \frac{\pi}{2} \log_e 2$$

Q83. The area of the region $A = \{(x, y) : |\cos x - \sin x| \leq y \leq \sin x, 0 \leq x \leq \frac{\pi}{2}\}$

$$(1) 1 - \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{5}}$$

$$(2) \sqrt{5} + 2\sqrt{2} - 4.5$$

$$(3) \frac{3}{\sqrt{5}} - \frac{3}{\sqrt{2}} + 1$$

$$(4) \sqrt{5} - 2\sqrt{2} + 1$$

Q84. Let $y = y(x)$ be the solution of the differential equation $x \log_e x \frac{dy}{dx} + y = x^2 \log_e x, (x > 1)$. If $y(2) = 2$, then $y(e)$ is equal to

$$(1) \frac{4+e^2}{4}$$

$$(2) \frac{1+e^2}{4}$$

$$(3) \frac{2+e^2}{2}$$

$$(4) \frac{1+e^2}{2}$$

Q85. If $\vec{a} = \hat{i} + 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$, $\vec{c} = 7\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{r} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$ then $\vec{r} \cdot \vec{c}$ is equal to:

$$(1) 34$$

$$(2) 12$$

$$(3) 36$$

$$(4) 30$$

Q86. Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b} = 3\hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{c}$ is a vector such that $\vec{c} \cdot (\vec{a} \times \vec{b}) + 25 = 0$, $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$

and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals:

$$(1) \frac{5}{\sqrt{2}}$$

$$(2) \frac{1}{5}$$

$$(3) \frac{1}{\sqrt{2}}$$

$$(4) \frac{3}{\sqrt{2}}$$

Q87. Shortest distance between the lines

$$\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5} \text{ and } \frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3} \text{ is}$$

(1) $2\sqrt{3}$
(3) $3\sqrt{3}$

(2) $4\sqrt{3}$
(4) $5\sqrt{3}$

Q88. The plane $2x - y + z = 4$ intersects the line segment joining the points $A(a, -2, 4)$ and $B(2, b, -3)$ at the point C in the ratio $2 : 1$ and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point $(a - b, b, 2b - a)$ then CP^2 is equal to:

(1) $\frac{17}{3}$
(3) $\frac{73}{3}$

(2) $\frac{16}{3}$
(4) $\frac{97}{3}$

Q89. If the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{1}$ and $\frac{x-a}{2} = \frac{y+2}{3} = \frac{z-3}{1}$ intersect at the point P , then the distance of the point P from the plane $z = a$ is :

(1) 16
(3) 10

(2) 28
(4) 22

Q90. Let $S = \{w_1, w_2, \dots\}$ be the sample space associated to a random experiment. Let $P(w_n) = \frac{P(w_{n-1})}{2}$, $n \geq 2$. Let $A = \{2k + 3l; k, l \in \mathbb{N}\}$ and $B = \{w_n; n \in A\}$. Then $P(B)$ is equal to

(1) $\frac{3}{32}$
(3) $\frac{1}{16}$

(2) $\frac{3}{64}$
(4) $\frac{1}{32}$